

Aufgabe 8.1: Quadraturgewichte bestimmen

a) Gegeben sind:

$$c_j := \frac{1}{b-a} \int_a^b \ell_{j,m}(x) dx, \int_a^b f(x) dx \approx (b-a) \sum_{j=0}^m c_j f(x_j)$$

i. $m = 1, a = 0, b = 1, x_0 = 0, x_1 = \frac{1}{2}$

$$\ell_{0,1}(x) = \frac{x - x_1}{x_0 - x_1} = \frac{x - \frac{1}{2}}{0 - \frac{1}{2}} = -2x + 1$$

$$\ell_{1,1}(x) = \frac{x - x_0}{x_1 - x_0} = \frac{x - 0}{\frac{1}{2} - 0} = 2x$$

$$c_0 = \int_0^1 \ell(x) dx = \int_0^1 (1 - 2x) dx = \left[x - x^2 \right]_0^1 = 0$$

$$c_1 = \int_0^1 \ell(x) dx = \int_0^1 2x dx = \left[x^2 \right]_0^1 = 1$$

$$\int_0^1 f(x) dx \approx (1 - 0) \sum_{j=0}^m c_j f(x_j) = \sum_{j=0}^1 c_j f(x_j) = 0 \cdot f(0) + 1 \cdot f\left(\frac{1}{2}\right)$$

ii. $m = 2, a = -1, b = 1, x_0 = -\frac{\sqrt{2}}{2}, x_1 = 0, x_2 = \frac{\sqrt{2}}{2}$

$$\ell_{0,2}(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} = x^2 - \frac{\sqrt{2}}{2}$$

$$\ell_{1,2}(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} = -2x^2 + 1$$

$$\ell_{2,2}(x) = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} = x^2 + \frac{\sqrt{2}}{2}$$

$$c_0 = \frac{1}{2} \int_{-1}^1 \ell(x) dx = \frac{1}{2} \int_{-1}^1 \left(x^2 - \frac{\sqrt{2}}{2} \right) dx = \left[\frac{1}{3} x^3 - \frac{\sqrt{2}}{4} x^2 \right]_{-1}^1 = \frac{1}{3}$$

$$c_1 = \frac{1}{2} \int_{-1}^1 \ell(x) dx = \frac{1}{2} \int_{-1}^1 (-2x^2 + 1) dx = \left[\frac{2}{3} x^3 + x \right]_{-1}^1 = \frac{1}{3}$$

$$c_2 = \frac{1}{2} \int_{-1}^1 \ell(x) dx = \frac{1}{2} \int_{-1}^1 \left(x^2 + \frac{\sqrt{2}}{2} \right) dx = \left[\frac{1}{3} x^3 + \frac{\sqrt{2}}{4} x^2 \right]_{-1}^1 = \frac{1}{3}$$

$$\int_{-1}^1 f(x) dx \approx 2 \sum_{j=0}^m c_j f(x_j) = 2 \sum_{j=0}^2 c_j f(x_j) = \frac{2}{3} \left(f\left(-\frac{\sqrt{2}}{2}\right) + f(0) + f\left(\frac{\sqrt{2}}{2}\right) \right)$$